INSTITUTO POLITÉCNICO NACIONAL

UNIDAD PROFESIONAL INTERDISIPLINARIA DE INGENIERÍA Y TECNOLOGÍAS AVANZADAS

CARRERA: INGENIERÍA TELEMÁTICA

UNIDAD DE APRENDIZAJE: MULTIMEDIA

Practica. Analisis de metadata de audios

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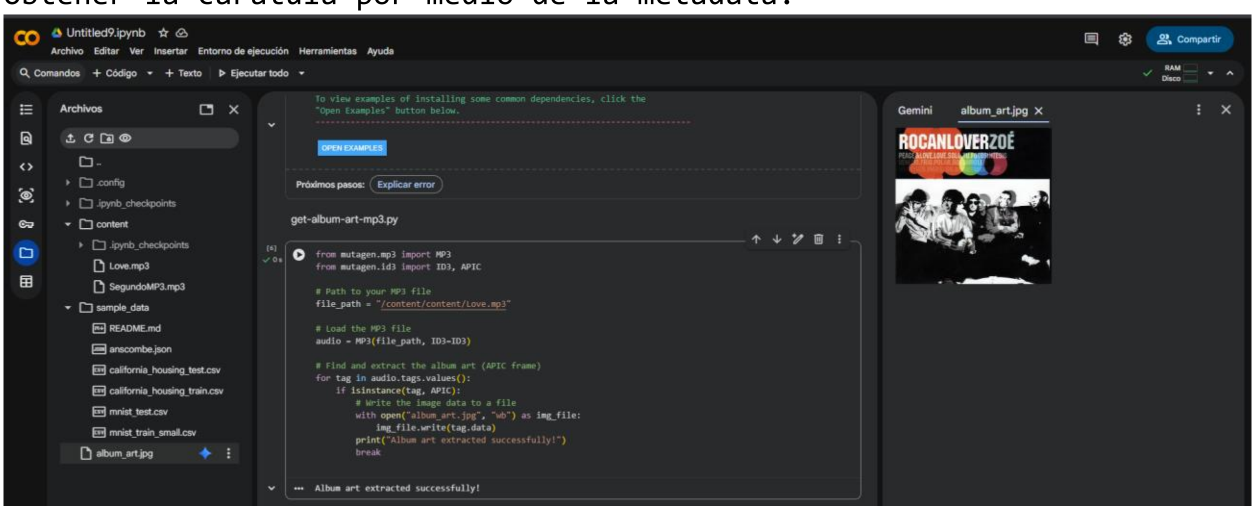
visualización de la atributos:

read-mp3-metadata.py

```python
!pip install mutagen

from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB

# Path to your MP3 file
file_path = "/content/content/Love.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Extract duration (in seconds)
duration = audio.info.length

# Extract ID3 tags (metadata)
tags = audio.tags

# Extract metadata if available
title = tags.get('TIT2', 'Unknown Title').text[0]  # Song title
artist = tags.get('TPE1', 'Unknown Artist').text[0]  # Artist name
album = tags.get('TALB', 'Unknown Album').text[0]  # Album name

# Print metadata
print(f"Title: {title}")
print(f"Artist: {artist}")
print(f"Album: {album}")
print(f"Duration: {duration:.2f} seconds")
```

```
Requirement already satisfied: mutagen in /usr/local/lib/python3.12/dist-packages (1.47.0)
Title: Love
Artist: Zoé
Album: Rocanlover
Duration: 203.68 seconds
```

Obtener la caratula por medio de la metadata:



Obtener la metadata:

get-metadata.py

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3

# Path to your MP3 file
-file_path = "path/to/your/file.mp3"
+file_path = "/content/content/Love.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Loop through all tags and print them
for tag in audio.tags:
    print(f"{tag}: {audio.tags[tag]}")
```

```
TIT2: Love
PRIV:WM/MediaClassPrimaryID:%}`Ñ#ãâK◻¡H¤*(D◻: PRIV(owner='WM/MediaClassPrimaryID', data=b'\xbc}`\xd1#\xe3\xe2
PRIV:WM/MediaClassSecondaryID:: PRIV(owner='WM/MediaClassSecondaryID', data=b'\x00\x00\x00\x00\x00\x00\x00\x0
TCON: Latin
TALB: Rocanlover
TPE2: Zoé
POPM:Windows Media Player 9 Series: POPM(email='Windows Media Player 9 Series', rating=255)
PRIV:PeakValue:¡ : PRIV(owner='PeakValue', data=b'\xa1\x7f\x00\x00')
PRIV:AverageLevel:{: PRIV(owner='AverageLevel', data=b'{\x1d\x00\x00')
TRCK: 2
TFLT: MPG/3
APIC:: APIC(encoding=<Encoding.LATIN1: 0>, mime='image/jpeg', type=<PictureType.OTHER: 0>, desc='', data=b'\x
APIC: : APIC(encoding=<Encoding.LATIN1: 0>, mime='image/jpeg', type=<PictureType.OTHER: 0>, desc=' ', data=b'
TCOM: León Larregui
TPE1: Zoé
TDRC: 2003
```

Cambio de los atributos en la metadata:

Change-mp3-metadata

```
from mutagen.id3 import ID3, TIT2, TPE1, TALB
from mutagen.mp3 import MP3

# Path to your MP3 file
file_path ="/content/content/Love.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Modify metadata (e.g., Title, Artist, Album)
audio.tags.add(TIT2(encoding=3, text="New Song Title"))   # Title
audio.tags.add(TPE1(encoding=3, text="New Artist Name"))  # Artist
audio.tags.add(TALB(encoding=3, text="New Album Name"))   # Album

# Save changes
audio.save()

print("Metadata updated successfully!")
```

```
Metadata updated successfully!
```

Volvemos a leer la metadata con la actualización de los atributos

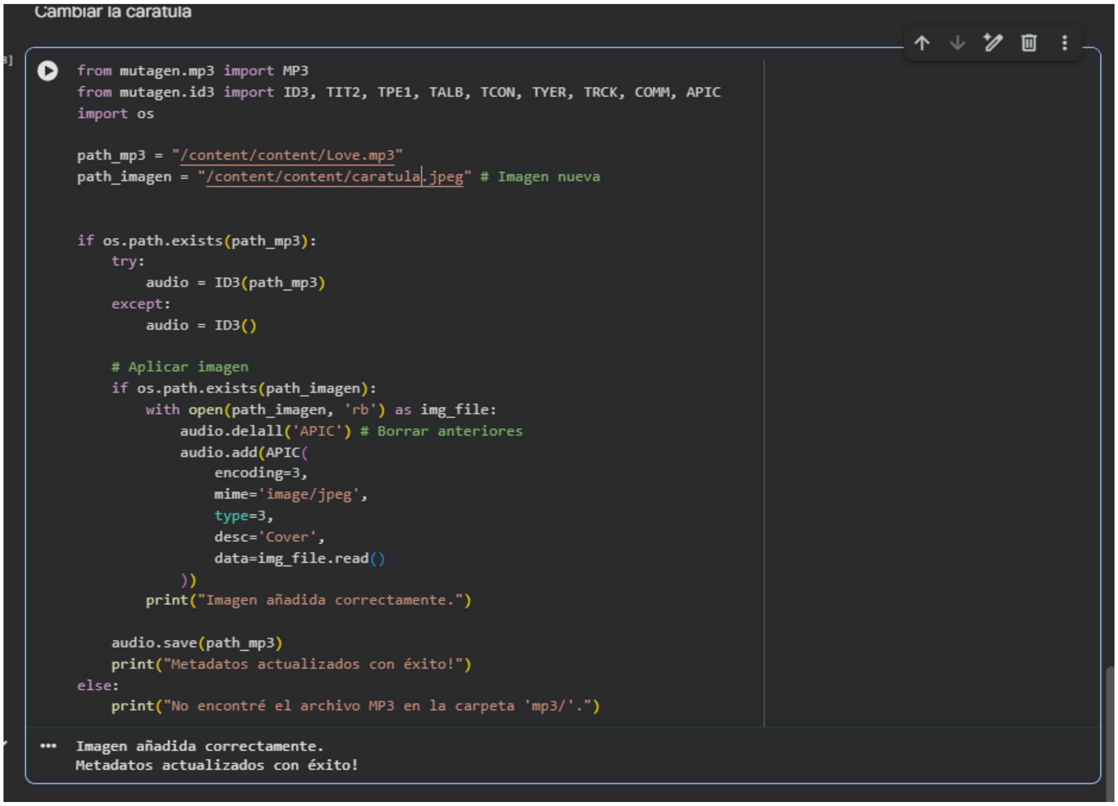
read-mp3-metadata.py

Gemini

```python
+!pip install mutagen
+
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB

# Path to your MP3 file
file_path = "/content/content/Love.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Extract duration (in seconds)
duration = audio.info.length

# Extract ID3 tags (metadata)
tags = audio.tags

# Extract metadata if available
title = tags.get('TIT2', 'Unknown Title').text[0]  # Song title
artist = tags.get('TPE1', 'Unknown Artist').text[0]  # Artist name
album = tags.get('TALB', 'Unknown Album').text[0]  # Album name

# Print metadata
print(f"Title: {title}")
print(f"Artist: {artist}")
print(f"Album: {album}")
print(f"Duration: {duration:.2f} seconds")
-
```

```
Requirement already satisfied: mutagen in /usr/local/lib/python3.12/dist-packages (1.47.0)
Title: New Song Title
Artist: New Artist Name
Album: New Album Name
Duration: 203.68 seconds
```

cambio de caratula:

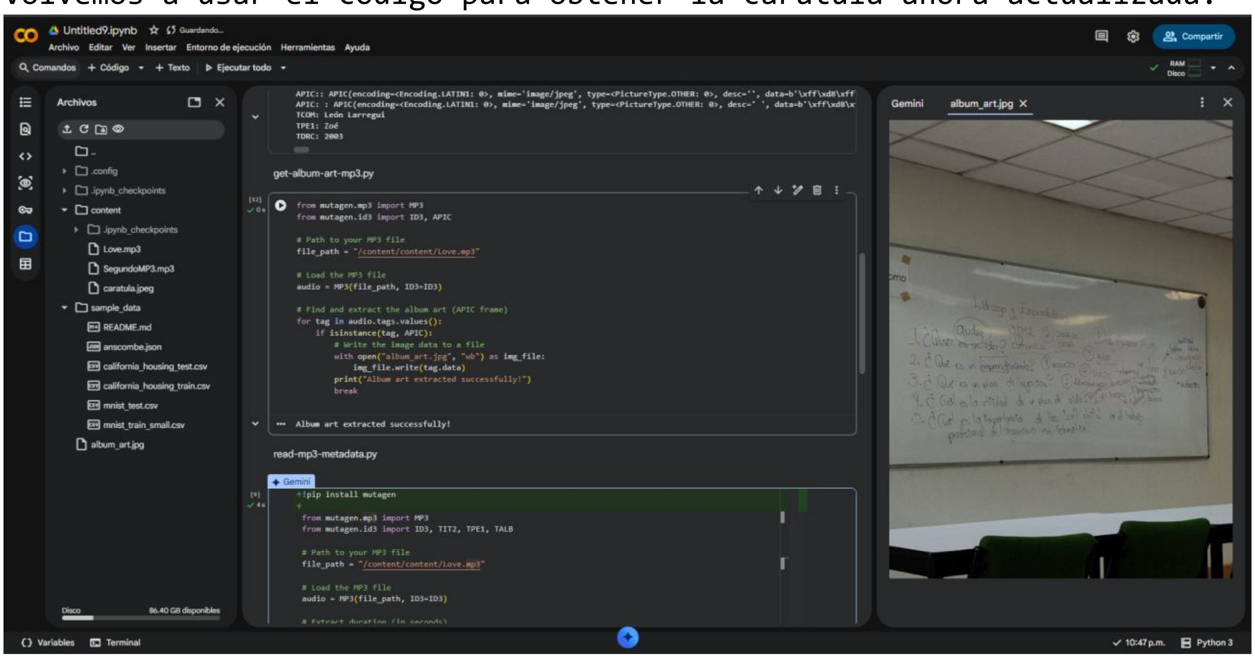


Cambiar la caratula

```python
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB, TCON, TYER, TRCK, COMM, APIC
import os

path_mp3 = "/content/content/Love.mp3"
path_imagen = "/content/content/caratula.jpeg" # Imagen nueva


if os.path.exists(path_mp3):
    try:
        audio = ID3(path_mp3)
    except:
        audio = ID3()

    # Aplicar imagen
    if os.path.exists(path_imagen):
        with open(path_imagen, 'rb') as img_file:
            audio.delall('APIC') # Borrar anteriores
            audio.add(APIC(
                encoding=3,
                mime='image/jpeg',
                type=3,
                desc='Cover',
                data=img_file.read()
            ))
        print("Imagen añadida correctamente.")

    audio.save(path_mp3)
    print("Metadatos actualizados con éxito!")
else:
    print("No encontré el archivo MP3 en la carpeta 'mp3/'.")
```

```
Imagen añadida correctamente.
Metadatos actualizados con éxito!
```

Volvemos a usar el código para obtener la caratula ahora actualizada:



Repetimos con un audio distinto:

Leer los atributos:

read-mp3-metadata.py

```
+!pip install mutagen
+
 from mutagen.mp3 import MP3
 from mutagen.id3 import ID3, TIT2, TPE1, TALB

 # Path to your MP3 file
-file_path = "/content/content/Love.mp3"
+file_path = "/content/content/SegundoMP3.mp3"

 # Load the MP3 file
 audio = MP3(file_path, ID3=ID3)

 # Extract duration (in seconds)
 duration = audio.info.length

 # Extract ID3 tags (metadata)
 tags = audio.tags

 # Extract metadata if available
 title = tags.get('TIT2', 'Unknown Title').text[0]  # Song title
 artist = tags.get('TPE1', 'Unknown Artist').text[0]  # Artist name
 album = tags.get('TALB', 'Unknown Album').text[0]  # Album name

 # Print metadata
 print(f"Title: {title}")
 print(f"Artist: {artist}")
 print(f"Album: {album}")
 print(f"Duration: {duration:.2f} seconds")
-
```

```
Requirement already satisfied: mutagen in /usr/local/lib/python3.12/dist-packages (1.47.0)
Title: Fly Me To The Moon
Artist: Frank Sinatra/ Count Basie And His Orchestra
Album: Nothing But The Best (2008 Remastered)
Duration: 147.52 seconds
```

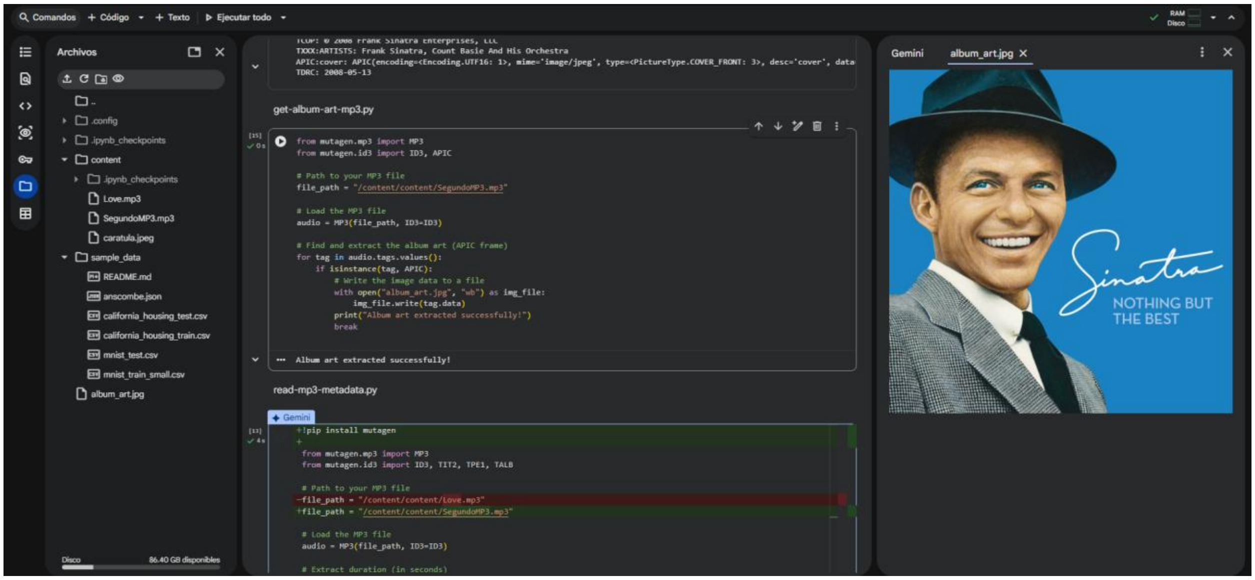
Leectura de la metadata:

get-metadata.py

```
 from mutagen.mp3 import MP3
 from mutagen.id3 import ID3

 # Path to your MP3 file
-file_path = "path/to/your/file.mp3"
+file_path = "/content/content/SegundoMP3.mp3"

 # Load the MP3 file
 audio = MP3(file_path, ID3=ID3)

 # Loop through all tags and print them
 for tag in audio.tags:
     print(f"{tag}: {audio.tags[tag]}")
```

```
TIT2: Fly Me To The Moon
TPE1: Frank Sinatra/ Count Basie And His Orchestra
TRCK: 8
TALB: Nothing But The Best (2008 Remastered)
TPOS: 1
TCON: Jazz
TLEN: 150000
TCOM: Bart Howard
TSRC: USRH10723029
TXXX:ITUNESADVISORY: 0
TXXX:BARCODE: 602537596379
TPUB: FRANK SINATRA DIGITAL REPRISE
TCOP: © 2008 Frank Sinatra Enterprises, LLC
TXXX:ARTISTS: Frank Sinatra, Count Basie And His Orchestra
APIC:cover: APIC(encoding=<Encoding.UTF16: 1>, mime='image/jpeg', type=<PictureType.COVER_FRONT: 3>, desc='cover', data
TDRC: 2008-05-13
```

Vista de caratula:



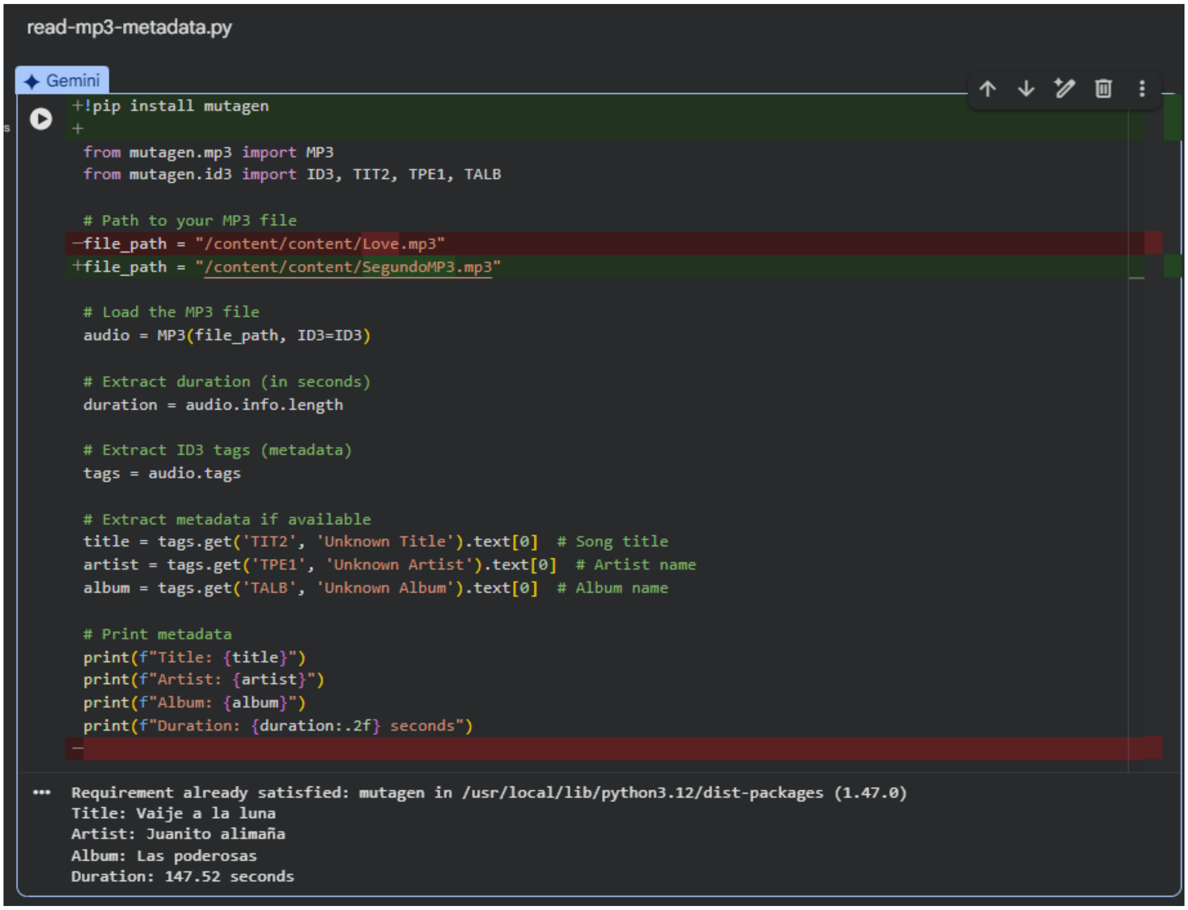
Cambio de atributos:

Change-mp3-metadata

```python
from mutagen.id3 import ID3, TIT2, TPE1, TALB
from mutagen.mp3 import MP3

# Path to your MP3 file
file_path ="/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Modify metadata (e.g., Title, Artist, Album)
audio.tags.add(TIT2(encoding=3, text="Vaije a la luna"))   # Title
audio.tags.add(TPE1(encoding=3, text="Juanito alimaña"))  # Artist
audio.tags.add(TALB(encoding=3, text="Las poderosas"))   # Album

# Save changes
audio.save()

print("Metadata updated successfully!")
```

```
Metadata updated successfully!
```

actualización de datos:

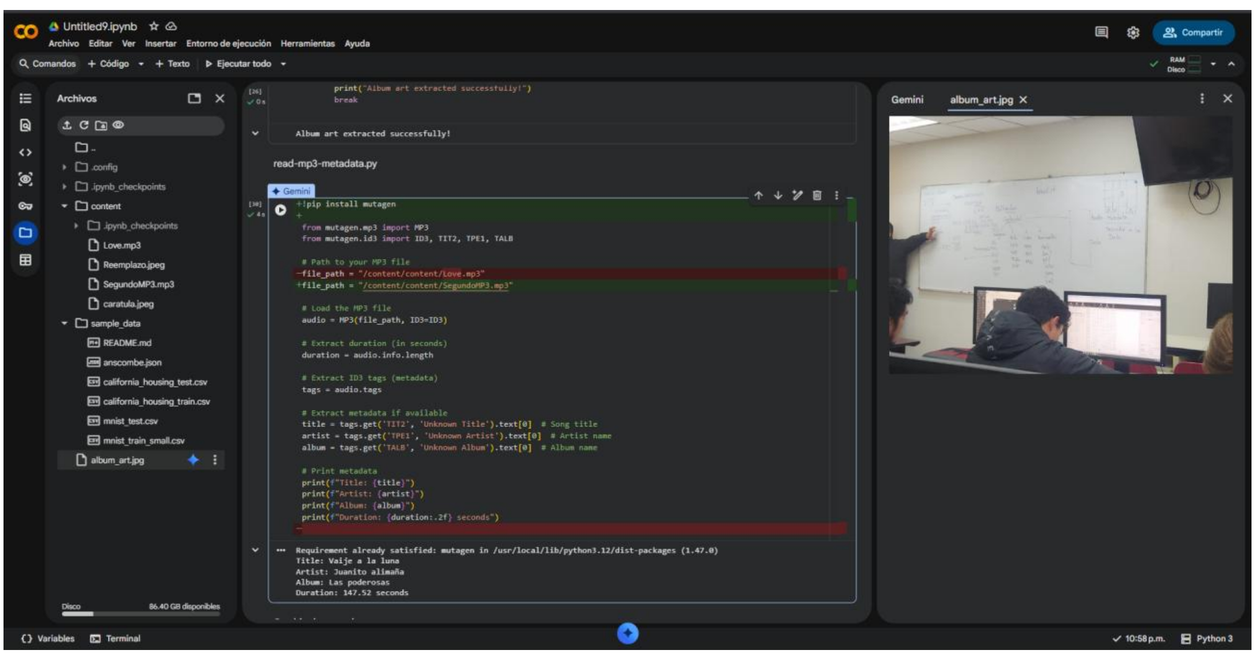


read-mp3-metadata.py

```
!pip install mutagen

from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB

# Path to your MP3 file
file_path = "/content/content/Love.mp3"
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Extract duration (in seconds)
duration = audio.info.length

# Extract ID3 tags (metadata)
tags = audio.tags

# Extract metadata if available
title = tags.get('TIT2', 'Unknown Title').text[0]  # Song title
artist = tags.get('TPE1', 'Unknown Artist').text[0]  # Artist name
album = tags.get('TALB', 'Unknown Album').text[0]  # Album name

# Print metadata
print(f"Title: {title}")
print(f"Artist: {artist}")
print(f"Album: {album}")
print(f"Duration: {duration:.2f} seconds")
```

```
Requirement already satisfied: mutagen in /usr/local/lib/python3.12/dist-packages (1.47.0)
Title: Vaije a la luna
Artist: Juanito alimaña
Album: Las poderosas
Duration: 147.52 seconds
```

Cambio de caratula:



códigos:

```
!pip install mutagen
```

cambio de la data:

```
from mutagen.id3 import ID3, TIT2, TPE1, TALB
from mutagen.mp3 import MP3

# Path to your MP3 file
file_path ="/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Modify metadata (e.g., Title, Artist, Album)
audio.tags.add(TIT2(encoding=3, text="Vaije a la luna"))   # Title
audio.tags.add(TPE1(encoding=3, text="Juanito alimaña"))  # Artist
audio.tags.add(TALB(encoding=3, text="Las poderosas"))   # Album

# Save changes
audio.save()

print("Metadata updated successfully!")
```

obtener metadata:

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3
```

```
# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Loop through all tags and print them
for tag in audio.tags:
    print(f"{tag}: {audio.tags[tag]}")
```

obtener caratula:

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, APIC

# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Find and extract the album art (APIC frame)
for tag in audio.tags.values():
    if isinstance(tag, APIC):
        # Write the image data to a file
        with open("album_art.jpg", "wb") as img_file:
            img_file.write(tag.data)
        print("Album art extracted successfully!")
        break
```

lectura metadata:

```
!pip install mutagen


from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB

# Path to your MP3 file
file_path = "/content/content/SegundoMP3.mp3"

# Load the MP3 file
audio = MP3(file_path, ID3=ID3)

# Extract duration (in seconds)
duration = audio.info.length

# Extract ID3 tags (metadata)
tags = audio.tags

# Extract metadata if available
title = tags.get('TIT2', 'Unknown Title').text[0]  # Song title
artist = tags.get('TPE1', 'Unknown Artist').text[0]  # Artist name
album = tags.get('TALB', 'Unknown Album').text[0]  # Album name

# Print metadata
print(f"Title: {title}")
```

```
print(f"Artist: {artist}")
print(f"Album: {album}")
print(f"Duration: {duration:.2f} seconds")
```

cambio de caratula:

```
from mutagen.mp3 import MP3
from mutagen.id3 import ID3, TIT2, TPE1, TALB, TCON, TYER, TRCK, COMM,
APIC
import os


path_mp3 = "/content/content/SegundoMP3.mp3"
path_imagen = "/content/content/Img2.jpeg" # Imagen nueva


if os.path.exists(path_mp3):
    try:
        audio = ID3(path_mp3)
    except:
        audio = ID3()

    # Aplicar imagen
    if os.path.exists(path_imagen):
        with open(path_imagen, 'rb') as img_file:
            audio.delall('APIC') # Borrar anteriores
            audio.add(APIC(
                encoding=3,
                mime='image/jpeg',
                type=3,
```

```
                desc='Cover',
                data=img_file.read()
            ))
        print("Imagen añadida correctamente.")

    audio.save(path_mp3)
    print("Metadatos actualizados con éxito!")
else:
    print("No encontré el archivo MP3 en la carpeta 'mp3/'.")
```